



IoT-Based Pedestrian Zone Safety System



TEAM MEMBERS:

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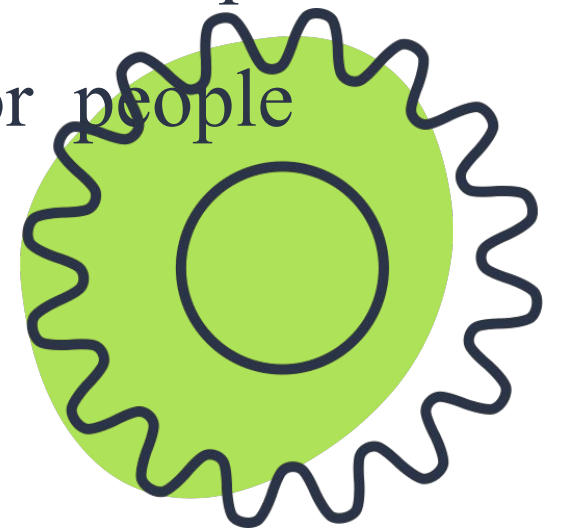
~MANJU R

~NANDHITHA S

PROBLEM

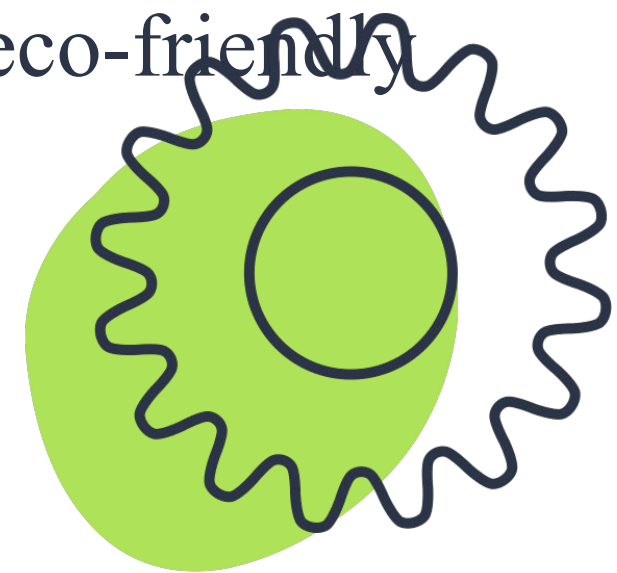
STATEMENT:

The IoT-Based Pedestrian Zone Safety System focuses on improving the safety of people walking in crowded city areas. It uses smart devices and sensors to detect if any vehicle (especially two-wheelers) is coming into the pedestrian zone where vehicles are not allowed. If a vehicle is detected, the system immediately sends alerts to both pedestrians and authorities and automatically stops the vehicle before it enters the pedestrian path. This system helps reduce accidents, keeps pedestrian areas safe, and makes cities more friendly for people walking.



INTRODUCTION:

Ensuring pedestrian safety in busy cities is challenging and needs smart solutions. The IoT-Based Pedestrian Zone Safety System uses ultrasonic, infrared, and RFID sensors, along with cameras. These detect unauthorized vehicles entering pedestrian-only areas. The system sends instant alerts to pedestrians and authorities. This helps prevent accidents and ensures quick action. It supports creating a safe, smart, and eco-friendly urban space.



ABSTRACT:

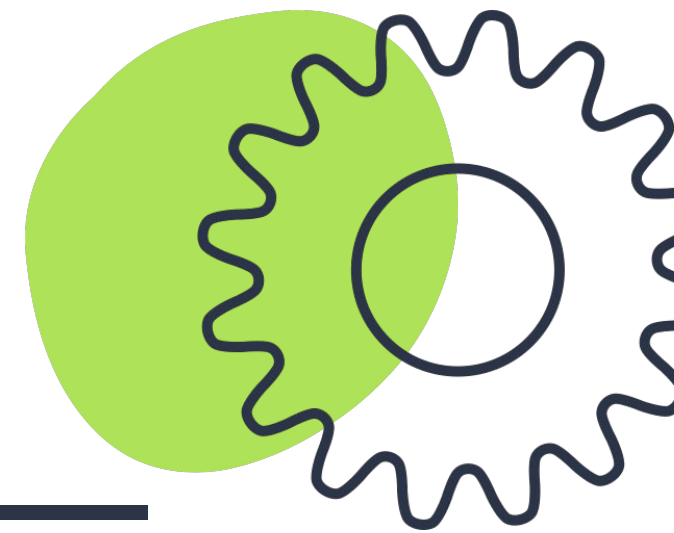
The IoT-Based Pedestrian Zone Safety System is designed to improve safety for people walking in cities using smart technology. It uses IoT sensors to detect when vehicles enter areas meant only for pedestrians. When this happens, the system sends real-time alerts to both pedestrians and authorities. In addition, the system can automatically stop two-wheelers from entering these restricted zones. This helps create a safe and well-organized environment for pedestrians. By reducing accidents and keeping pedestrian areas secure, this system supports a smarter, safer, and more sustainable urban infrastructure, especially in busy city areas.



SOLUTION:

Our working model uses a camera feed processed by YOLOv8 object detection in Python.

When a vehicle — like a bicycle or motorcycle — enters the detection zone, the system calculates the distance from the gate and immediately sends a command to close the gate via our controller. When a human enters, the system also calculates the distance but does not close the gate, allowing safe pedestrian access. All detections are logged in real-time to our Firebase cloud database, making it easy to monitor and store activity records.



COMPONENTS TO BE USED:

1. Iot-enabled sensors

Optical sensor & ultrasonic sensors – to detect the vehicles

Pir sensors – to detect the human movement

2. Motion sensors

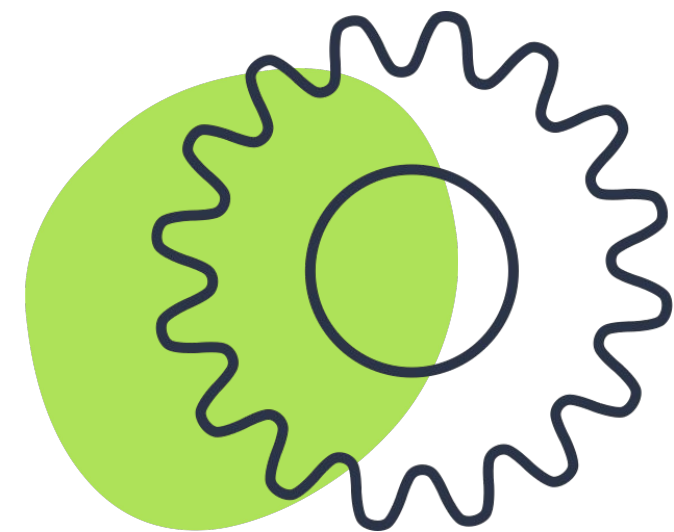
3. Processing unit - (arduino)

4. Alert systems

Led displays

5. Power supply

6.ESP32camera



USE OF COMPONENTS:

1. Optical sensor & ultrasonic sensors:

These sensors help find and track vehicles that enter pedestrian areas.

2. PIR sensors:

They sense human movement, helping to keep track of people walking in the area.

4. Motion sensors:

They sense movement of both vehicles and people in the area.

5. Processing unit (Arduino):

The Arduino collects information from all the sensors and controls the whole system based on that data.



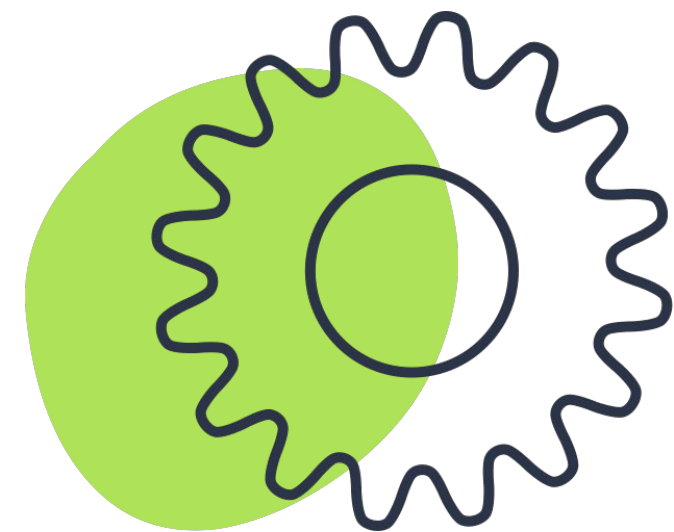
USE OF COMPONENTS:

6. Power supply:

This gives electricity to all the components so the system works all the time.

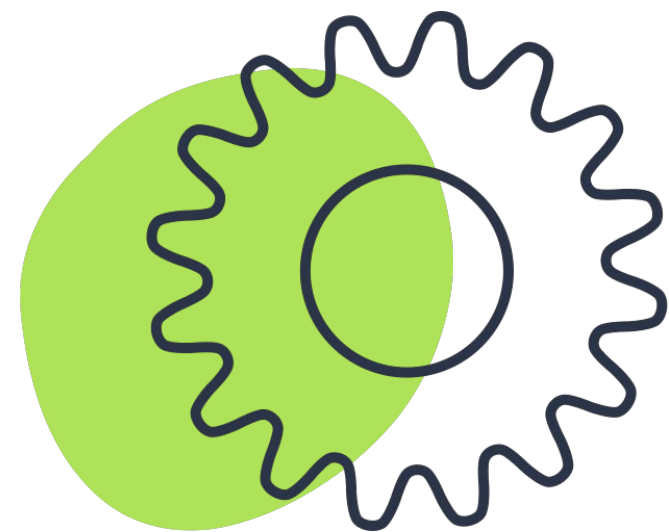
7. LED display:

The screen will show live warnings and messages to tell people when a vehicle enters the pedestrian zone.

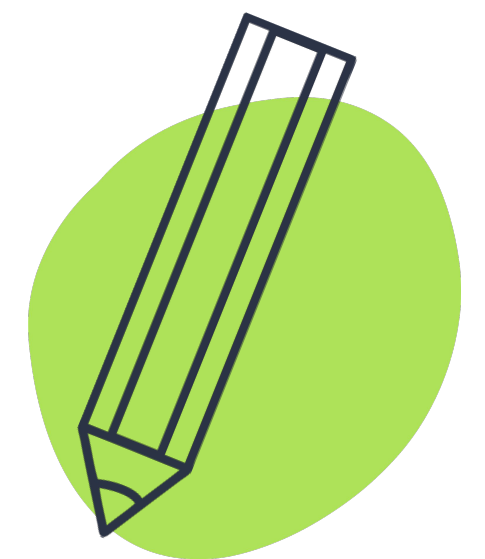
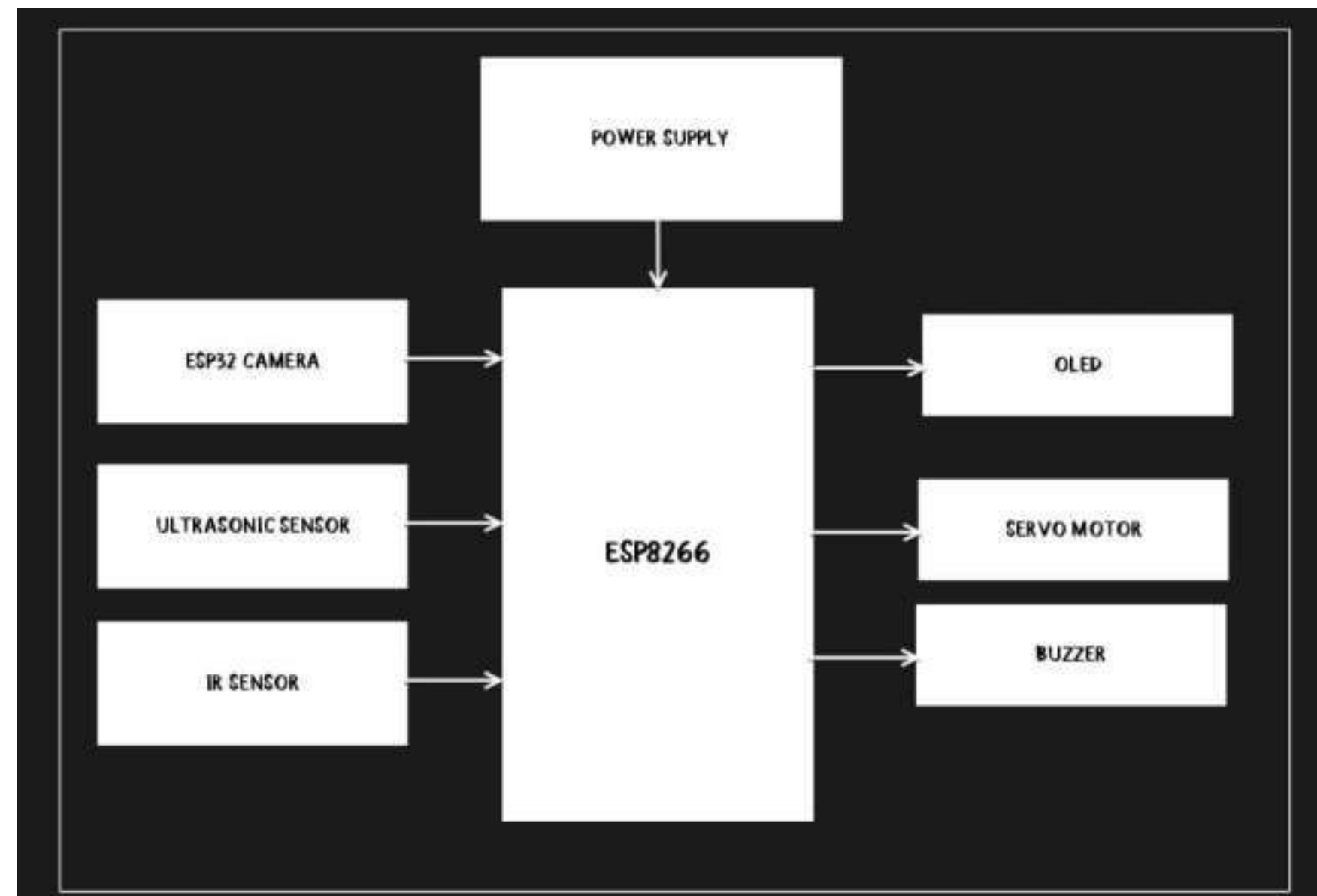


ADVANTAGES:

- ~Keeps pedestrians safe by stopping vehicles from entering walking areas.
- ~Gives instant alerts to people and authorities so they can take quick action.
- ~Helps reduce accidents in crowded city areas by using smart technology.
- ~Creates a better and safer space for people to walk.
- ~Encourages drivers to follow rules and avoid restricted zones.



BLOCK DIAGRAM:



Impact and Social Relevance:

Our IoT-Based Pedestrian Zone Safety System significantly enhances pedestrian safety by preventing unauthorized vehicle entry in real time, reducing the risk of accidents in busy areas. By automating gate control with intelligent object detection, it eliminates the need for costly and error-prone manual monitoring.

The system promotes safer urban spaces, supports traffic regulation enforcement, and provides transparent activity logs for authorities through cloud storage. This contributes to building smarter, safer cities that prioritize pedestrian well-being and reduce road hazards.



COLCLUSION:

The IoT-based pedestrian zone safety system is a smart and useful way to keep people safe in city areas.

It uses sensors and real-time alerts to stop vehicles from entering walking zones, which helps reduce accidents.

This system makes the area more organized and safer for people to walk.

Overall, it helps create a pedestrian-friendly environment and improves safety and movement in cities.



Thank you!

